

Course Proposal

Course Code	To be assigned (3xx level)
Department	CSE
Course Name	Software Development using AI
Credits	4
Course Offered to	UG (B.Tech. CSE, year 3/4)
Course Description	Software Development using AI examines AI-native software development where developers collaborate with coding agents. The course introduces the orchestration of agent skills and the Model Context Protocol, enabling students to configure agents that satisfy most test cases. It covers translating natural-language requirements into executable specifications, iterative loop engineering with trace analysis, and evaluation of automated testing and reliability frameworks. Throughout the 13-week term, students engage in lectures, labs, and a substantial project that integrates multi-module development using emerging AI tools. By the end of the course, students will be able to design, implement, and assess AI-assisted development workflows. The course does not become a general introduction to AI or machine learning.

Pre-requisites

Pre-requisite (Mandatory)	Pre-requisite (Desirable)	Pre-requisite (Other)
CSE201	None	None

Post Conditions (Course Outcomes)

CO	Students will be able to...
CO1	Configure coding agents and agent skills, using an agent orchestration framework and Model Context Protocol, such that at least 80% of the provided test cases pass.
CO2	Construct executable specifications from a given natural-language requirements document, using spec-driven development tools, covering every use case in the supplied requirements document.
CO3	Analyze execution traces produced by coding agents during iterative loop engineering, identifying at least three distinct failure causes with supporting log evidence.
CO4	Evaluate automated testing and reliability frameworks for agent-generated code, justifying each evaluation decision against at least one stated requirement.
CO5	Evaluate a complete AI-native software development workflow for a given multi-module project, demonstrated live with two scenario walkthroughs chosen by the evaluator.

Weekly Lecture Plan

Week Number	Lecture Topic	COs Met	Tutorial	Assignments / Project
1	- AI-assisted coding and developer workflows - Requirements to executable specifications	CO1, CO2	Trace and analyze 2 sample developer workflow diagrams in groups, identifying tool components using standard terminology.	
2	- Requirements to executable specifications - Spec-driven development tools	CO2	Worked trace-diagnosis of 2 natural-language requirements documents to define functional use cases and boundaries in pairs.	
3	- Advanced spec-driven workflows - Agent skills and tools configuration	CO1, CO2	Group troubleshooting session mapping agent skill configurations against supplied orchestration framework errors.	Quiz 1
4	- Agent skills and tools configuration - Model Context Protocol bindings	CO1	Peer review of Model Context Protocol binding configurations for edge-case test setups.	Assignment 1 released
5	- Model Context Protocol bindings - Execution log inspection and warnings	CO1, CO3	Trace-diagnosis exercise inspecting execution logs from iterative agent loops to isolate warning messages.	Assignment 1 due Assignment 2 released

Week Number	Lecture Topic	COs Met	Tutorial	Assignments / Project
6	• Execution log inspection and warnings • Iterative development and trace debugging	CO3	Walkthrough and discussion of trace error outputs from agent loop engineering, matching logs to failure causes.	
7	• Advanced trace debugging practice • Automated testing and reliability frameworks comparison	CO3, CO4	Mid-semester review problem-solving session comparing automated testing framework approaches.	Assignment 2 due Mid-sem exam
8	• Automated testing and reliability frameworks comparison • Usability and evaluation of test coverage reports	CO4	Comparative analysis exercise contrasting automated testing and reliability frameworks across common metrics in teams.	
9	• Usability and evaluation of test coverage reports • Coverage report evaluation and feedback	CO4	Worked evaluation of test coverage reports against stated requirements in pairs.	Report 1 released Project Milestone 1 released
10	• Hands-on coding and multi-module project design • Multi-module project integration practice	CO5	Architectural design review of multi-module project interface specifications in small groups.	Report 1 due
11	• Multi-module project integration practice • Emerging developer tools and live build evaluation	CO5	Peer evaluation of multi-module project integration checkpoints and interface compliance.	Project Milestone 1 due Project (final) released
12	• Live build evaluation exercises	CO5	Mock presentation and scenario walkthrough preparation for final build artifact evaluations.	
13	Project work and review			Project (final) due

Weekly Lab Plan

Week Number	Laboratory Exercise	COs Met	Platform (Hardware / Software)
1	Introduction to AI-Assisted Developer Workflows: Explore basic AI-assisted coding tools and examine developer environment configurations in pairs.	CO1	Student development machines, software development environments
2	Spec-Driven Development Tools Practice: Construct executable test specifications from natural-language requirements using spec-driven development tools.	CO2	Student development machines, software development environments
3	Agent Skills and Tool Configuration: Configure coding agents and agent skills using an agent orchestration framework on local test environments.	CO1	Student development machines, access to AI coding tools/APIs
4	Model Context Protocol Bindings: Configure Model Context Protocol bindings for coding agents using an agent orchestration framework for 3 test configurations.	CO1	Student development machines, access to AI coding tools/APIs
5	Advanced MCP Bindings and Log Inspection: Refine MCP bindings and practice examining execution logs produced by coding agents during iterative loops.	CO1, CO3	Student development machines, software development environments
6	Iterative Development and Trace Debugging: Analyze trace error outputs produced by coding agents during iterative loop engineering to identify failure causes.	CO3	Student development machines, software development environments
7	Advanced Trace Debugging Practice: Conduct advanced trace debugging practice with complex failure logs and multi-step agent traces.	CO3	Student development machines, software development environments
8	Post-Midsem Catch-up and Open Lab: Open lab session to catch up on debugging exercises and review automated testing frameworks.	CO3	Student development machines

Week Number	Laboratory Exercise	COs Met	Platform (Hardware / Software)
9	Test Coverage and Evaluation Reports: Evaluate test coverage reports from automated testing frameworks and draft justifications against requirements.	CO4	Student development machines, software development environments
10	Multi-Module Project Design: Design agentic tools, specifications, and loops into a unified multi-module software project structure.	CO5	Student development machines, software development environments
11	Multi-Module Project Integration Practice: Integrate multi-module project components and evaluate emerging developer tools in live builds.	CO5	Student development machines, access to AI coding tools/APIs
12	Live Build Evaluation Exercises: Conduct live build evaluations of AI-native software development workflows with scenario walkthroughs.	CO5	Student development machines, access to AI coding tools/APIs

Assessment Plan

Type of Evaluation	% Contribution in Grade
Quiz	10
Assignment (2)	20
Mid-sem	20
Report	15
Project (2)	35
Total	100

Resource Material

Type	Title
Reference (reading)	Andrii Tkachuk (2026). SPECIFICATION-DRIVEN DEVELOPMENT AS STRUCTURED APPROACH FOR SOURCE CODE GENERATION. DOI: 10.52058/2786-6025-2026-4(58)-2631-2644
Reference (reading)	Kevin Lano, Qiaomu Xue (2022). Code Generation by Example. DOI: 10.5220/0010973600003119
Reference (reading)	Anthony Shaw, Amin Beheshti (2026). Complexity Backpressure for AI Coding Agents: A Cross-Model Study on Substantive Software Tasks. DOI: 10.1109/sse72781.2026.00047
Reference (reading)	Jahnvi Bellapukonda (2026). A Comparative Evaluation of LLM-based Coding Agents for Automated Software development Tasks. DOI: 10.2139/ssrn.6755658
Reference (reading)	Mingxuan Xiao, Yan Xiao, Shunhui Ji, Jiahe Tu (2026). BASFuzz: Toward Robustness Evaluation of LLM-based NLP Software via Automated Fuzz Testing. DOI: 10.1145/3843770

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